Software Requirements Specification

Functional Requirements

1. User Registration and Authentication

The system shall allow users to securely register and authenticate in the platform. The system shall support the following features:

• Users shall be able to **register and log in** using email and password.
• Users shall be able to **log in using Google Sign-In**.
• Users shall be able to **reset passwords using email verification**.
• Users shall be able to **log out securely from the system**.

The system shall support the following user roles:

• **Gig Professional** – users who provide services or complete tasks.
• **Client** – users who post tasks and hire gig professionals.
• **Manager** – users who manage tasks and approve hiring decisions within organizations.
• **Root Client (Organization Owner)** – users who can manage managers and oversee tasks.

2. Profile Management

2.1 Gig Professional Profile

The system shall allow gig professionals to create and manage their professional profiles. The system shall allow gig professionals to:

• Create and update their professional profile.
• Add skills, expertise, and tools they use.
• Upload portfolio items or previous work samples.
• Upload a profile picture.
• Add completed tasks to their profile as work experience.

2.2 Client and Manager Profiles

The system shall allow clients and managers to:

• Create and update their profile information.
• Manage account details and organization information.

3. Gig Service Creation

The system shall allow gig professionals to publish services they offer. The system shall allow gig professionals to:

• Create service listings describing their skills and services.
• Add service titles, descriptions, categories, and pricing information.
• Upload sample work or portfolio images.
• Edit or remove previously published services.

Published services shall be visible to clients in the **gig marketplace**.

4. Task Posting

The system shall allow clients and root clients to create and manage tasks. The system shall allow users to:

• Create a task with title, description, required skills, budget, and deadline.
• Attach files or references related to the task.
• Edit or delete tasks before hiring a gig professional.
• View all posted tasks in their dashboard.

Tasks shall be visible to gig professionals in the **task marketplace**.

5. Task Discovery and Proposal Submission

The system shall allow gig professionals to discover and apply for tasks. The system shall allow gig professionals to:

• Browse available tasks posted by clients.
• Search tasks using keywords.
• Filter tasks by category, required skills, budget, and deadline.
• View full task descriptions.
• Submit proposals with a message and relevant portfolio items.

Clients shall be notified when a new proposal is submitted.

6. Creator Selection and Hiring

The system shall allow clients and managers to review proposals and select suitable gig professionals. The system shall allow them to:

• View all proposals submitted for a task.
• Review creator profiles and portfolio.
• Accept or reject proposals.
• Send hiring requests to selected gig professionals.

Managers shall be able to **approve or reject hiring requests** before the task officially begins.

7. Task Assignment and Management

The system shall allow the manager to manage and monitor active tasks. The system shall allow managers to:

• Assign tasks to approved gig professionals.
• Track the status of tasks (Active, Submitted, Completed).
• Monitor task progress and updates.

Clients and managers shall be able to view **real-time task progress updates**.

8. Deliverable Submission

The system shall allow gig professionals to submit deliverables after completing assigned work. The system shall allow gig professionals to:

• Upload deliverable files.
• Add comments or descriptions explaining the work.
• Submit final deliverables for review.

Clients and managers shall receive notifications when a deliverable is submitted.

9. Deliverable Review and Approval

The system shall allow clients and managers to review deliverables submitted by gig professionals. The system shall allow them to:

• Review uploaded deliverable files.
• Request revisions if the work does not meet requirements.
• Approve the deliverable if the work meets expectations.

Once approved, the task status shall change to **Completed**.

10. Payments and Reviews

The system shall allow clients to complete payment and provide feedback after task completion. The system shall allow:

• Clients to initiate payment for completed tasks.
• Secure payment processing through a payment gateway.
• Clients to leave ratings and reviews for gig professionals.

Non-Functional Requirements

1. Performance Requirements

- The system shall respond to user requests (page loads, searches, contract viewing) within **3 seconds** under normal conditions.
- The system shall support processing of at least **200 gig searches per minute** without delay.
- The system shall handle simultaneous gig posting and proposal submissions without affecting response time.

2. Reliability Requirements

- The system shall maintain an uptime of **99% or higher** during operational hours.
- The system shall ensure that contract and payment **data are not lost** in the event of system failure.
- The system shall automatically **retry failed transactions** related to work submissions and payments.

3. Scalability Requirements

- The system shall support up to **5,000 concurrent users** without performance degradation.
- The platform shall allow future addition of more **gig categories and users** without system redesign.

4. Availability Requirements

- The system shall be **available 24/7**, except during scheduled maintenance periods.
- Scheduled **maintenance** downtime shall not exceed **2 hours per month**.

5. Usability Requirements

- **New users** shall be able to complete the **gig posting or service requesting process** within **5 minutes**.
- The system shall provide **clear error message**s when invalid data is entered.

- The user interface shall be responsive and usable on **mobile and desktop browsers.**

6. Security Requirements

- The system shall use **multi-factor authentication** for all user logins.
- The system shall implement **role-based authorization** for Clients, Gig Workers, and Managers.
- All payment-related data shall be transmitted using **HTTPS encryption.**

7. Maintainability Requirements

- The system shall be developed using a **modular architecture** to allow future updates.
- Code shall follow **standard coding** practices and documentation.
- **System logs shall be maintained** to help in debugging and issue tracking.

8. Efficiency Requirements

- The system shall **not exceed 70% CPU utilization** during peak usage.
- Database operations such as **gig search and profile retrieval shall be optimized** to avoid resource wastage.

9. Portability Requirements

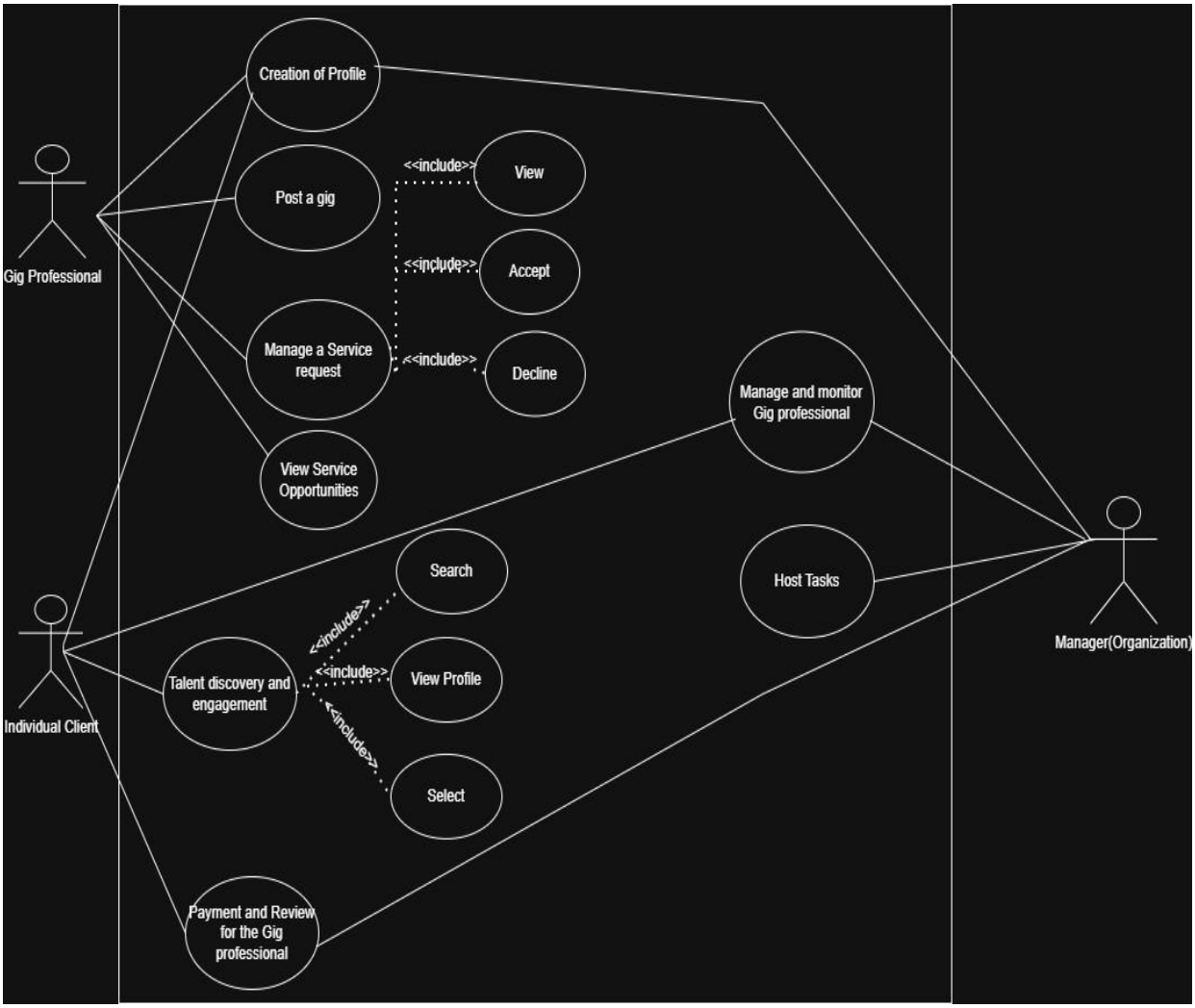
- The application shall run on **Windows, macOS, and Linux servers**.
- The web platform shall function correctly on **major browsers (Chrome, Firefox, Edge, Safari)**.

10. Reusability Requirements

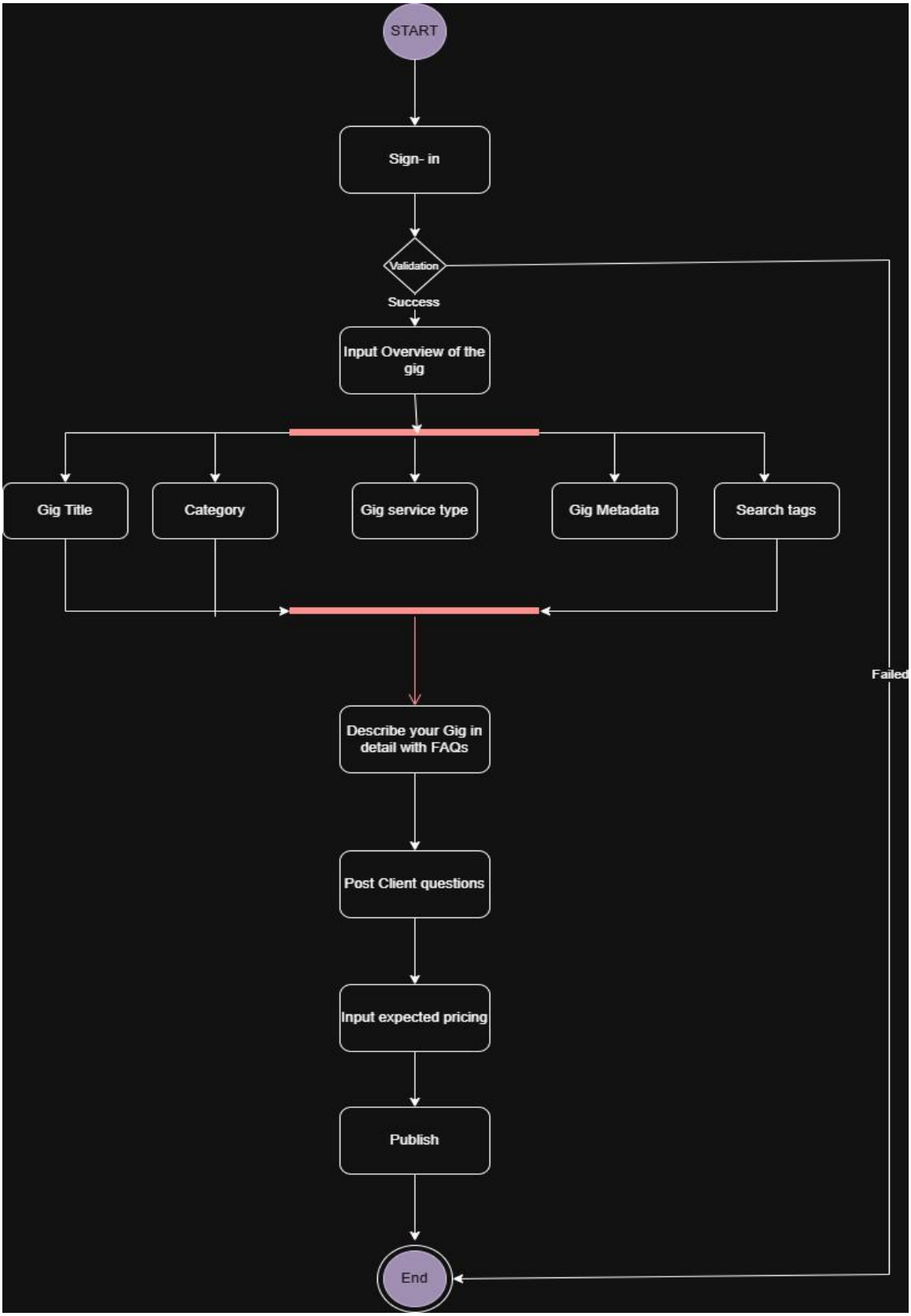
- The user **authentication** module shall be **reusable** for future web modules.
- The **payment processing** component shall be designed for reuse in other transaction-based features.

UML Diagrams

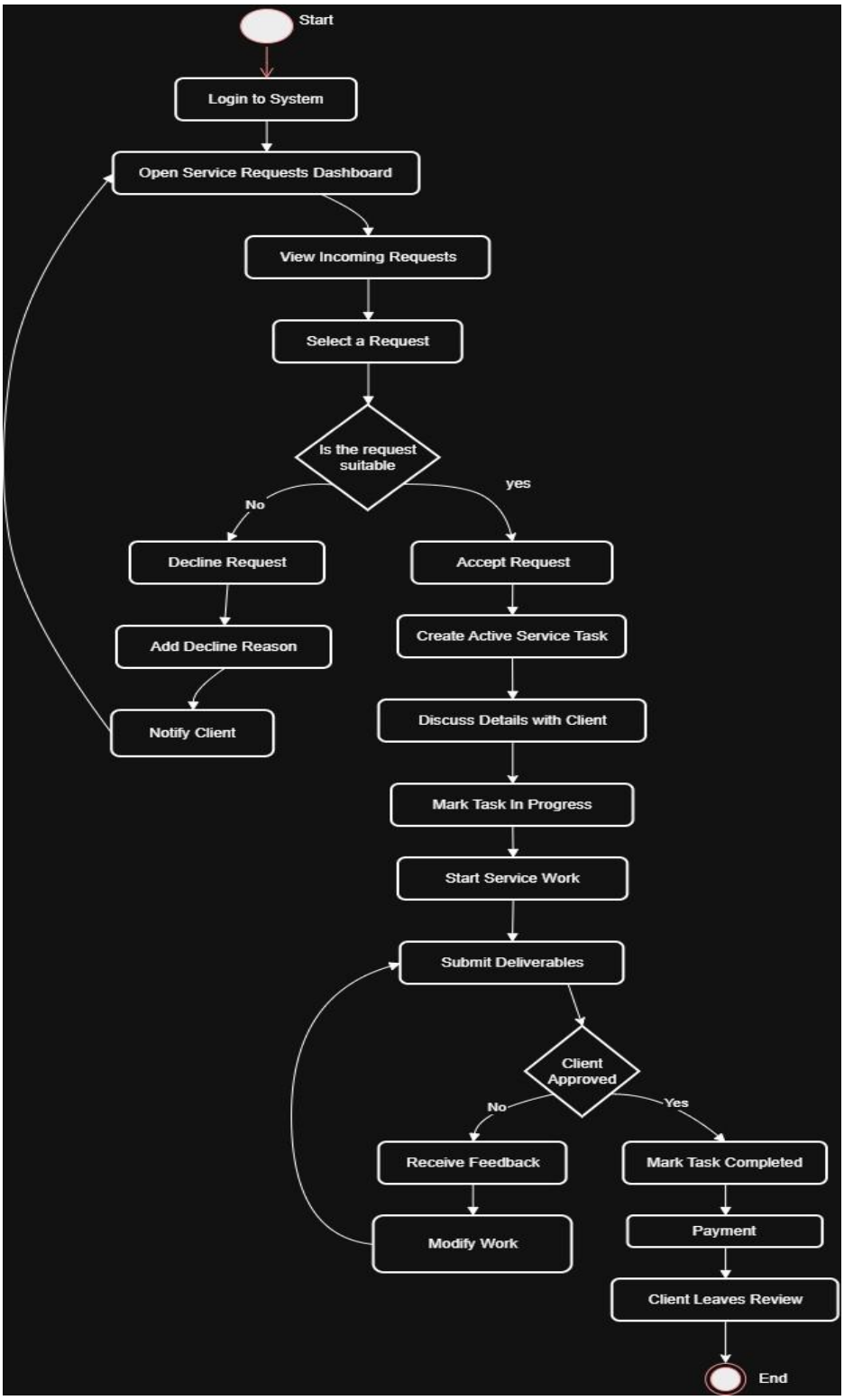
1. Use Case Diagram



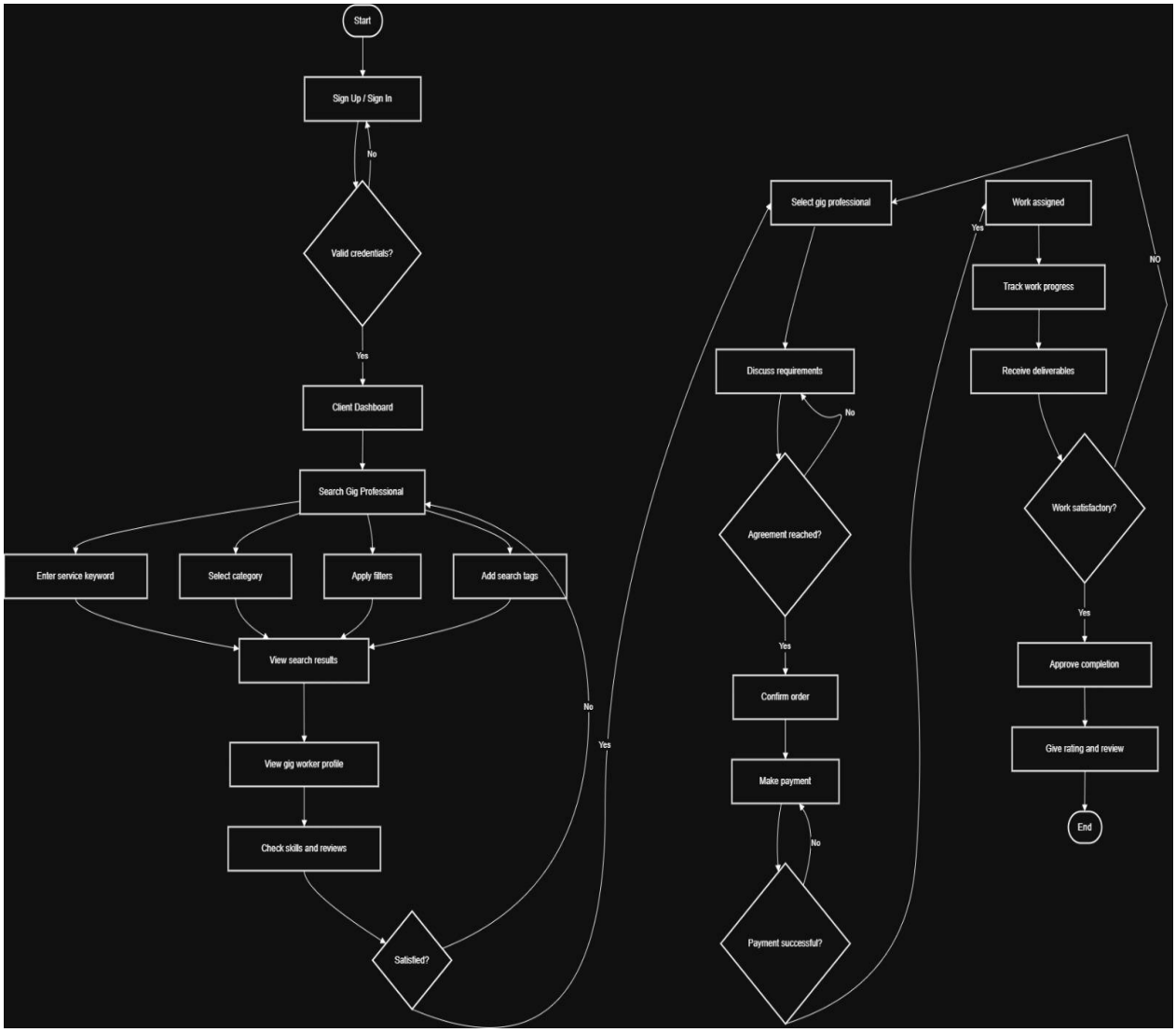
2. Activity Diagrams • Post a gig



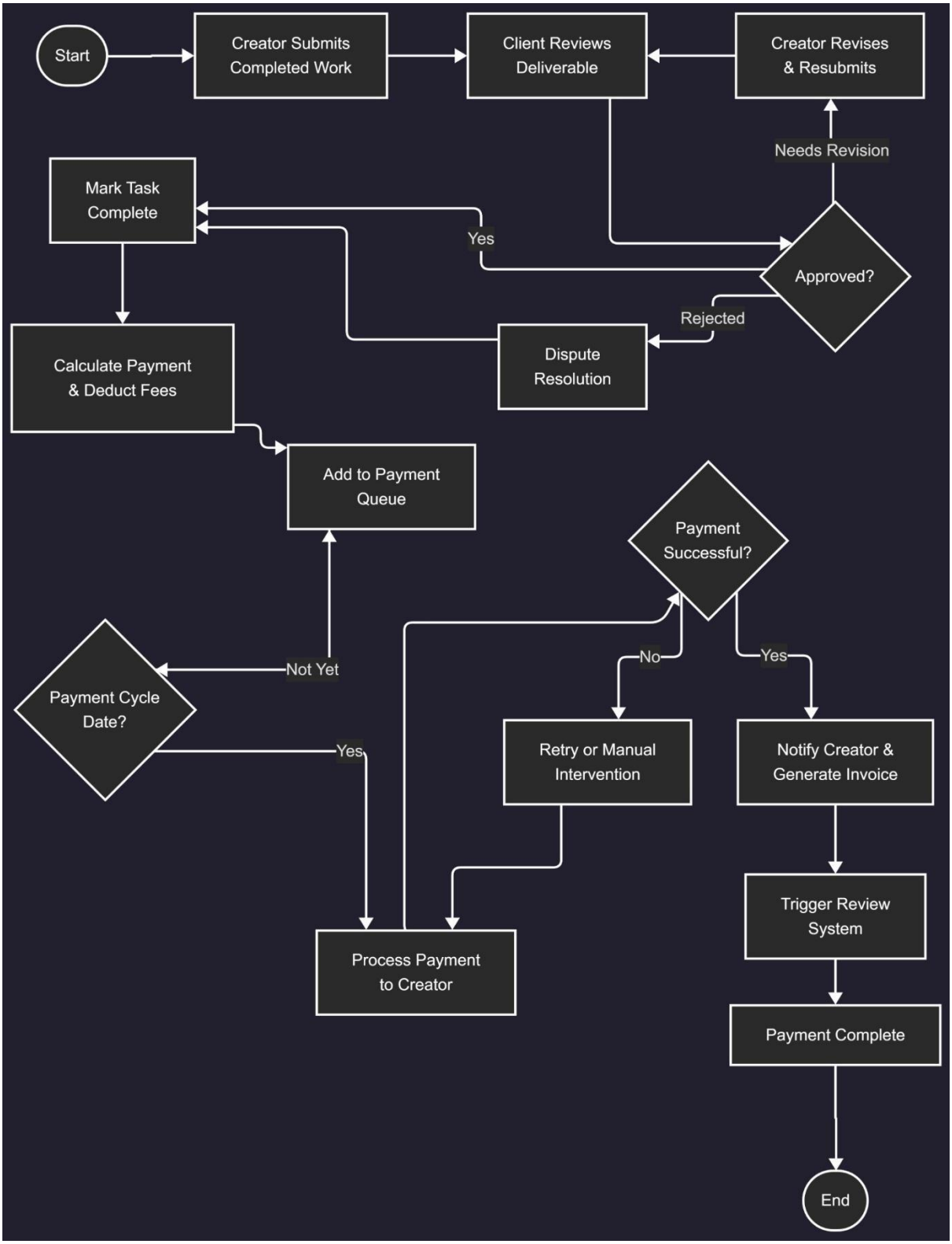
- Manage a service request



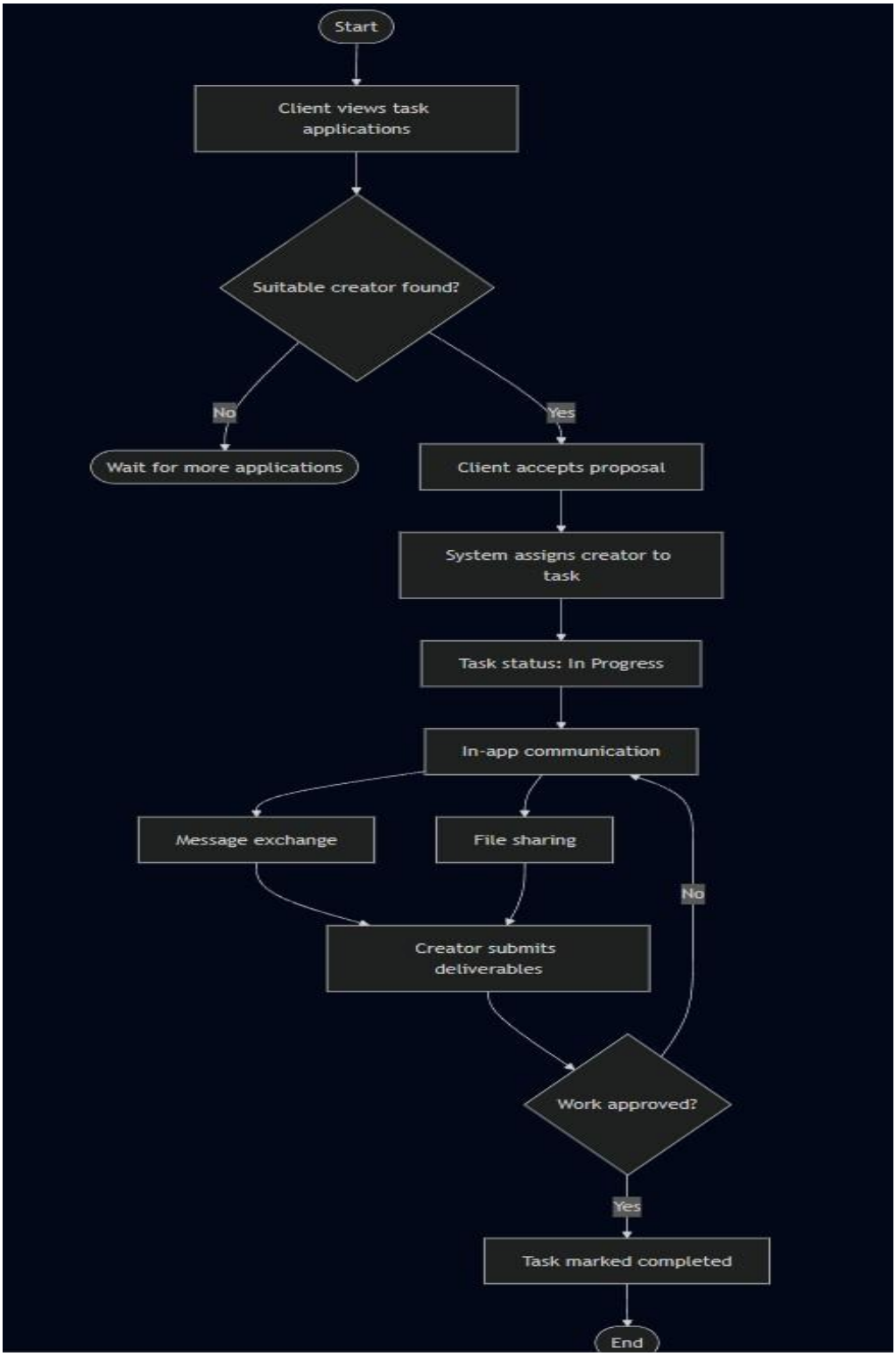
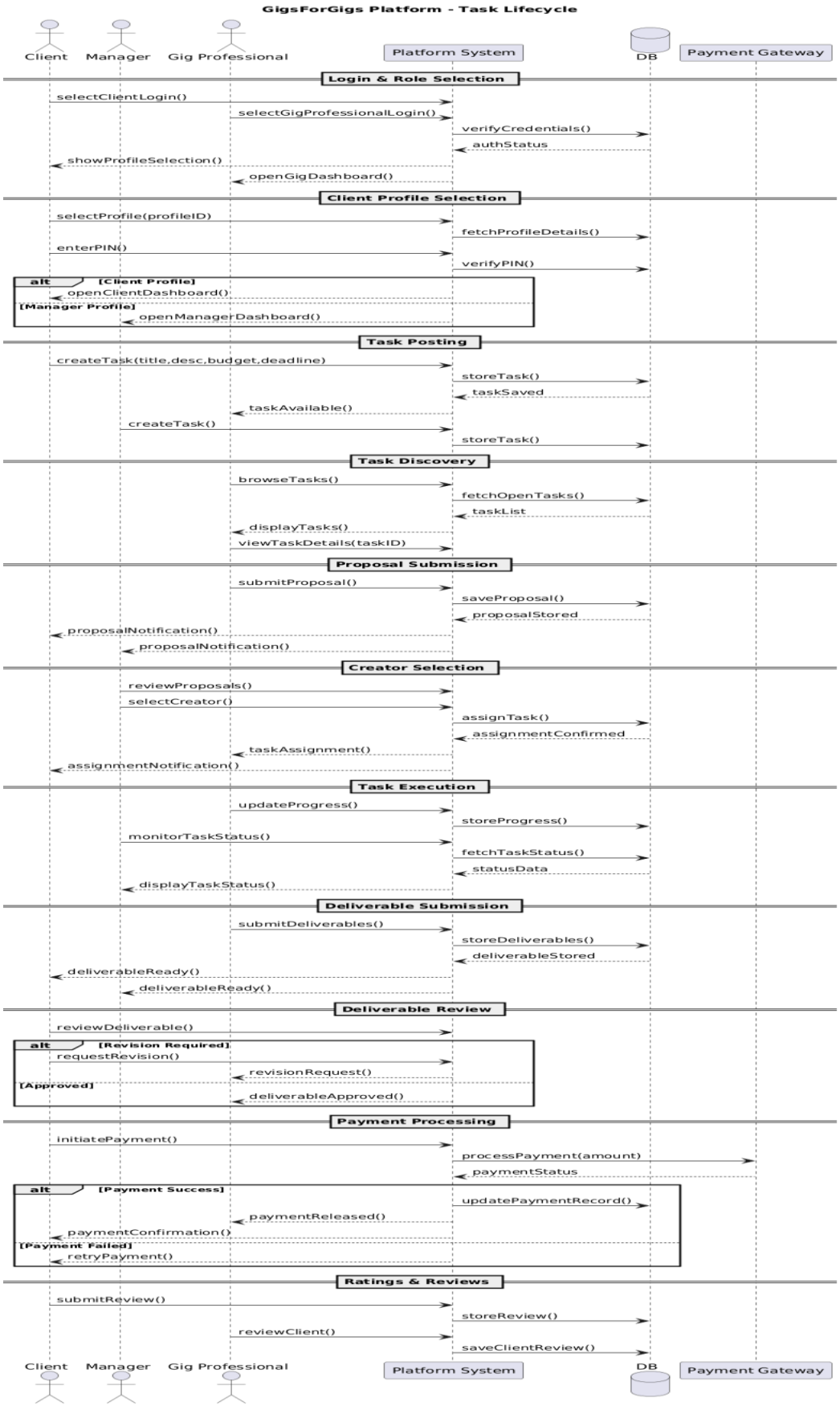
- Talent discovery and engagement



- Payment and Review the gig professional



- Monitor and manage the gig professional



- **Sequence Diagram**